

OKORO EMMANUEL ONYEDIKACHI

Full-Stack & Frontend Software Engineer

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PROFESSIONAL SUMMARY

Detail-oriented Full-Stack Developer with hands-on experience building scalable, secure, and user-centric web applications. Currently contributing to Terapage — a live research activity management platform — where I own frontend architecture across multiple complex activity types and collaborate directly with designers and the backend team. Expert in the MERN stack and Next.js (App Router), with a strong focus on robust authentication, performance-optimized frontends, and RESTful APIs. Passionate about software architecture, type safety, accessibility, and delivering polished, production-grade experiences.

PROFESSIONAL EXPERIENCE

Frontend / Full-Stack Developer | [Terapage](#)

2026 – Present

Terapage is a research activity management platform serving Researchers, Collaborators, Observers, and Participants. Stack: React (Vite), Node.js/Express, MongoDB/Mongoose, TypeScript.

- Owned end-to-end frontend development for six distinct activity types — Document Review, Poll & Survey, Upload Audio/Video/Image, and Multi-Task — translating Figma designs into production-quality React components.
- Engineered a pixel-accurate annotation system for the Document Review activity, resolving complex positioning bugs caused by dynamic scroll offsets and PDF viewport scaling.
- Implemented persistent multilingual support (i18next) across all activity components, including language state hydration on page reload for Participant-facing views.
- Built a global terminology rename (Project → Study, Activity → Research Activity) spanning frontend components, backend routes, Mongoose schemas, notification templates, and transactional emails — ensuring zero regressions across the platform.
- Developed the Participant-side Multi-Task module with search, pagination, draft visibility, and progress indicators, resolving tab-styling and navigation edge cases across role contexts.
- Resolved a suite of media player bugs across Upload Audio, Video, and Image activities including subtitle sync, AI indicator display, annotate flow, and download functionality.
- Collaborated directly with the backend team to align API contracts, debug Mongoose aggregation pipelines, and integrate MongoDB Schema.Types.Mixed fields for flexible activity data models.
- Maintained strict ESLint compliance throughout (no unused vars, correct useEffect deps, no console.log, no var) and authored structured PR descriptions covering root cause, changes, and behavior verification.
- Initiated the Terapage React Native mobile app for the Participant experience using Expo, TypeScript, Expo Router v3, Zustand, TanStack Query, and NativeWind — currently in active development.

Full-Stack & Frontend Developer | [Personal & Practice Projects](#)

2022 – 2026

Full-Stack Next.js Authentication System *Next.js 15 · TypeScript · MongoDB · JWT · Bcryptjs · Nodemailer*

- Architected a secure authentication boilerplate using Next.js App Router with user registration, login, and profile management.
- Implemented Bcrypt password hashing and JWT management via HTTP-only cookies to mitigate XSS and CSRF risks.
- Engineered a Middleware gatekeeper for centralized route protection between public routes and private dashboards.
- Developed a Mongoose Singleton pattern to optimize DB connections and prevent OverwriteModelError during hot-reloads.

Country Quiz Master *React · TypeScript · Tailwind CSS 4.0 · Vite*

- Built an ultra-fast geography quiz app with 100 dynamic questions covering global flags and capitals.
- Engineered a Smart Scoring engine using TypeScript `Record<number, number>` state mapping to prevent double-scoring.

- Architected strict type safety via custom interfaces, eliminating UI breaks caused by malformed API data.

MERN Exercise Tracker *MongoDB Atlas · Express · React · Node.js*

- Developed a full-stack CRUD app for exercise log management, using Vite for high-performance frontend builds.
- Engineered a RESTful API with Node.js/Express and schema-based data modeling via Mongoose and MongoDB Atlas.

Jumia E-commerce Interface & Weather Dashboard *React · Redux · Tailwind CSS · Open Metro API*

- Built a fully responsive Jumia frontend clone using Atomic Design principles and reusable React components.
- Implemented a persistent Dark/Light theme engine using Redux and localStorage.
- Validated calculation logic with component-level tests using Jest and Vitest.

CORE TECHNICAL SKILLS

Frontend:	Next.js 14/15 (App Router), React.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, Vite
Mobile:	React Native, Expo (SDK 51+), Expo Router v3, NativeWind, TypeScript
State & UI:	Zustand, TanStack Query, Redux, React Hooks, Tailwind CSS 4.0, Styled Components, Bootstrap
Backend:	Node.js, Express.js, RESTful API Design, JWT, Bcryptjs, Nodemailer, Zod
Database:	MongoDB Atlas, Mongoose (ODM), MySQL, Schema.Types.Mixed, aggregation pipelines
Tooling:	Git/GitHub, Postman, Axios, ESLint, Prettier, i18next
Testing:	Jest, Vitest (Unit & Component Testing)
Soft Skills:	Problem-solving, Technical Debugging, Cross-functional Collaboration, Adaptability

EDUCATION

B.Sc. Chemistry — Second Class Honours (Upper Division)

2017 – 2023

University of Lagos

PROJECTS & PORTFOLIO

- Next.js Auth System — Secure full-stack auth with JWT & Middleware. [Github](#) | [Live App](#)
- Country Quiz Master — High-performance React/TS quiz with smart scoring. [Github](#) | [Live App](#)
- Weather Dashboard — Responsive React app with persistent theme engine. [Github](#) | [Live App](#)
- Jumia Replica — Functional React frontend replicating complex e-commerce UI. [github.com](#)
- MERN Exercise Tracker — Full-stack CRUD app for exercise logging. [github.com](#)
- Terapage Mobile (In Progress) — React Native Participant app using Expo + Expo Router v3.